

Current-code pipeline table: qml2.0/codes

Scope: the 37 audited computational stages, formatted in the eight-column structure of the supplied PDF. This is a code-only record; formulas describe repository operations, with external-library boundaries identified. No code or datasets were modified. Source key: I = initialdata.py; D = descriptor.py; R = run1.py; C = circuit.py; T = train.py; P = plot\_run1\_results.py. All source locations refer to the completed audit of 24 September 2026. Energies are Hartree unless stated. Important findings: default H2S data fails the current AO-selection contract; FE\_prime is missing from the plotting CLI; descriptor production lacks explicit convergence acceptance. Numerical checks and limitations remain in the accompanying audit report.

| Layer                    | Input                                                                                | Exact method / function                                                                        | What is done                                                                                                                                                                                                                                                                                   | Output                                     | Basis                    | Formula                                                                                                                                                                                                                            | Important element definition in formula                                                                   |
|--------------------------|--------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| 1 Geometry               | Molecule specification; optimized reference; stretch tangent; Hessian; atomic masses | I optimize_equilibrium771-824; StretchPath194-261; calculate_harmonic431-579; scan_grid858-865 | Cartesian coordinates; mass-weighted proper alignment follows the raw stretch. The scan includes both endpoints and contains 30 geometries. Harmonic constants use consistent SI quantities before conversion to angstrom.                                                                     | 30 aligned geometries; scan coordinate     | Cartesian                | $k = \sum_{a,b,x,y} t_{ax} H_{abxy} t_{by}, \quad \mu = \sum_{a,x} m_a t_{ax}^2, \quad \omega = \sqrt{k/\mu},$ $\Delta = \sqrt{\frac{3\hbar}{\mu\omega}}, \quad s_j = s_e - \Delta + \frac{2j\Delta}{29}, \quad j = 0, \dots, 29.$ | a,b: atoms; x,y: Cartesian components; j=0,...,29. Geometry in angstrom; SI used for harmonic conversion. |
| 2 AO overlap             | Geometry; basis; charge; spin                                                        | I rhf_arrays887-893; D calculate_record144-161                                                 | Full spherical-AO overlap from the molecular geometry, basis, charge and spin. Integral evaluation is delegated to PySCF. Dimensionless matrix; no active restriction here.                                                                                                                    | S: M by M                                  | Full AO                  | $S \in \mathbb{R}^{M \times M}.$                                                                                                                                                                                                   | M: full spatial AO count. Dimensionless overlap; PySCF integral boundary.                                 |
| 3 RHF reference          | Molecule; SCF settings; saved density initial guess for descriptor RHF               | I configured_rhf607-613, polish_orbitals616-647, solve_point660-743; D139-159                  | Initial-data RHF checks convergence and internal stability. Descriptor RHF is recomputed using the saved AO density as an initial guess. Density is spin summed; the Fock matrix is in Hartree. SCF and integral internals remain library boundaries.                                          | P,F,C: M by M; RHF energy                  | AO / MO                  | $P = C \operatorname{diag}(n_p) C^{\mathsf{T}}, \quad F = h + J[P] - \tfrac{1}{2} K[P].$                                                                                                                                           | n_p: spatial orbital occupation. P spin summed; F and total energy in Ha.                                 |
| 4 Core/space definitions | Molecule registry; AO labels; shell metadata                                         | I ACTIVE_SPACES37-41; D select_valence_aos91-119                                               | FCI frozen MOs and excluded descriptor AOs define separate restrictions. MP2 and RDM construction freeze no orbitals. Descriptor exclusion uses AO/shell metadata; H2S retains 10 AOs.                                                                                                         | Frozen-MO specification; retained AO map v | MO / active AO           | Excluded descriptor AO: $Z > 2 \wedge$ principal shell = 1.<br>Descriptor MP2: frozen = 0.                                                                                                                                         | FCI frozen-core definition is distinct from the descriptor AO exclusion. No MP2 frozen core.              |
| 5 MP2 amplitudes         | Recomputed canonical RHF reference                                                   | D calculate_record156-169                                                                      | Restricted MP2 uses all occupied and virtual spatial MOs. Amplitude and energy construction is delegated to PySCF; saved-array amplitude checks agree to roundoff. No unaudited library-internal amplitude formula is asserted.                                                                | Restricted doubles t2                      | Full MO                  | $t_2 \in \mathbb{R}^{n_{\text{occ}} \times n_{\text{occ}} \times n_{\text{virt}} \times n_{\text{virt}}}.$                                                                                                                         | Occupied and virtual axes are spatial orbitals. Amplitudes dimensionless; PySCF boundary.                 |
| 6 Used RDM construction  | Reference determinant; zero singles; restricted t2                                   | D build_consistent_rdms60-88                                                                   | The determinant reference and first-order correction use zero singles and restricted doubles. Apply the following algebra separately to the one- and two-particle RDMs. Determinant conversion and RDM APIs are external boundaries; normalized-state diagnostics do not replace this algebra. | Consistent 1-RDM and 2-RDM                 | Determinant / MO         | $w = \langle \chi   \chi \rangle, \quad R_0 = R[\Phi],$ $R_1 = R[\Phi + \chi] - R[\Phi] - R[\chi], \quad R_2 = R[\chi] - w R_0,$ $R = R_0 + R_1 + R_2.$                                                                            | R denotes either RDM API; Phi is reference, chi is correction; w is its squared norm.                     |
| 7 Tensor ordering        | Raw rank-four RDM-derived array                                                      | D transform_rank456-57; extract_descriptors129-136                                             | Full rank-four arrays preserve raw PySCF index ordering. Only triple-descriptor extraction performs the displayed axis permutation. There is no explicit four-spin-block pipeline.                                                                                                             | Raw array; extraction-only permutation     | AO / Lowdin AO           | $\Lambda_{pqrs} \xrightarrow{\text{transpose}(0,2,1,3)} \tilde{\Lambda}_{pqrs}.$                                                                                                                                                   | Permutation swaps axes 1 and 2 (zero based). It is not a global change to stored raw tensors.             |
| 8 MO→AO                  | MO RDMs; full real coefficient matrix C                                              | D calculate_record171-175; transform_rank456-57                                                | MO-to-AO transformations use real arrays and ordinary transpose, without conjugation. Every axis spans the full space.                                                                                                                                                                         | Full AO 1-RDM and 2-RDM                    | MO to AO                 | $D_{\text{AO}} = C D_{\text{MO}} C^{\mathsf{T}},$ $G_{abcd} = \sum_{pqrs} C_{ap} C_{bq} C_{cr} C_{ds} G_{pqrs}.$                                                                                                                   | a,b,c,d: AO indices; p,q,r,s: MO indices. All axes run over M.                                            |
| 9 AO restriction         | Full transformed arrays; source and canonical AO maps                                | D122-136,182-186; R ao_selection140-172                                                        | Select AOs only after full-space transformations. The producer saves source-ascending indices; the loader rebuilds canonical atom/AO order. Local retained indices and full source indices are distinct.                                                                                       | Selected arrays in canonical order         | Full to active Lowdin AO | $v_i : i \mapsto \text{full source AO index}, \quad A_{ij}^{\text{sel}} = A_{v_i v_j}.$ $v_{\text{BeH}_2, \text{model}} = [5, 1, 2, 3, 4, 6] \quad (\text{zero-based}).$                                                           | v_i maps local i to full source AO. BeH2 displayed source indices are zero based.                         |

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|---------------------------------|-----------------------------------------------------|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 10 Disconnected reference       | Full RHF AO density P                               | D disconnected_rhf47-48;176                                | Disconnected reference uses the full RHF spin-summed density. The displayed indices are free indices, not summation indices.                                                                                                                                              | Disconnected G0: M to fourth power         | Full AO                     | $(G_0)_{pqrs} = P_{pq}P_{rs} - \frac{1}{2}P_{ps}P_{rq}.$                                                                                                                                                                                                                                                                                                               | p,q,r,s are free indices. Factor 1/2 follows the implemented spin-summed expression.  |
| 11 Reference subtraction        | Consistent AO 2-RDM; RHF disconnected reference     | D176-181                                                   | The reference subtraction is performed in the full AO space. The conventional subtraction using correlated density is separately stored and is not the descriptor input.                                                                                                  | Lambda: M to fourth power                  | Full AO                     | $\Lambda = G_{\text{consistent}} - G_0[P_{\text{RHF}}], \quad \Lambda \in \mathbb{R}^{M \times M \times M \times M}.$                                                                                                                                                                                                                                                  | Not the separately stored conventional cumulant based on correlated density.          |
| 12 Extra diagnostic contraction | Stored RDM diagnostics                              | D160-210                                                   | No additional all-ones contraction of the fourth axis is executed. Raw RDMs, order components and the alternate cumulant are stored diagnostics. There is no additional formula for an absent stage.                                                                      | No extra all-ones contraction              | Stored diagnostics          | No additional repository formula; see operation and boundary.                                                                                                                                                                                                                                                                                                          | No corresponding downstream stage is executed.                                        |
| 13 FCI and target               | Canonical RHF; configured core and active electrons | I frozen_core_fci896-943; calculate_point946-990; R227-232 | Canonical-RHF CASCI defines the active determinant space. Active one-electron and pair-packed two-electron integrals are saved but not consumed by descriptor production. Core energy includes nuclear and frozen-core terms. CASCI solving remains an external boundary. | FCI energy; CI; target y; active integrals | Active MO / determinant     | $p \in [n_{\text{core}}, M), \quad n_{\text{cas}} = M - n_{\text{core}}, \quad N_{\text{cas}} = N_e - 2n_{\text{core}},$<br>$E_{\text{FCI}} = E_{\text{active}} + E_{\text{core}}, \quad y = E_{\text{FCI}} - E_{\text{RHF,input}},$<br>$\text{shape}(\text{CI}) = \left( \binom{n_{\text{cas}}}{N_{\text{cas}}/2}, \binom{n_{\text{cas}}}{N_{\text{cas}}/2} \right).$ | ncore: frozen spatial MOs; active interval excludes M. Target is in Ha.               |
| 14 Overlap powers               | Full overlap S                                      | D lowdin_factors51-53; R208-211                            | Full-space overlap powers precede selection. The producer applies no threshold or clipping. The loader requires the minimum eigenvalue to exceed the stated threshold and checks matrix identities.                                                                       | H=S^(1/2); J=S^(-1/2)                      | Full AO                     | $S = U \text{diag}(s)U^{\text{T}}, \quad H = U \text{diag}(\sqrt{s})U^{\text{T}},$<br>$J = U \text{diag}(s^{-1/2})U^{\text{T}}, \quad \min(s) > 10^{-8} \text{ (loader)}.$                                                                                                                                                                                             | Here H is overlap square root, not stage-1 Hessian; s contains overlap eigenvalues.   |
| 15 One-body Löwdin              | H,J; full RHF density; AO Fock matrix               | D182-184; R212-213                                         | Full Löwdin-AO matrices. No explicit transpose averaging occurs at this stage. The density is the RHF density, not the correlated density.                                                                                                                                | P_L,F_L: M by M                            | Full Lowdin AO              | $P_L = HP_{\text{RHF}}H, \quad F_L = JF_{\text{AO}}J, \quad P_L, F_L \in \mathbb{R}^{M \times M}.$                                                                                                                                                                                                                                                                     | P_L dimensionless; F_L in Ha. No explicit symmetrization.                             |
| 16 Rank-four Löwdin             | H; full AO Lambda                                   | D56-57,182-184; R216                                       | All four axes are transformed in the full space before selecting AOs. The same overlap square root is used on every axis.                                                                                                                                                 | Lambda_L: M to fourth power                | Full Lowdin AO              | $(\Lambda_L)_{abcd} = \sum_{pqrs} H_{ap}H_{bq}H_{cr}H_{ds}(\Lambda_{\text{AO}})_{pqrs}.$                                                                                                                                                                                                                                                                               | a,b,c,d,p,q,r,s each span full space. All four factors are H.                         |
| 17 T extraction                 | Lambda_L; retained AO map v                         | D extract_descriptors122-136                               | The repeated local index is not contracted. A full triple array and its strict-triple vector are saved. Current Run 1 constructs no triple input and does not consume these arrays.                                                                                       | T_full; strict-triple vector               | Active Lowdin AO            | $T_{ijk} = (\Lambda_L)_{v_i v_k v_j v_k} \quad (\text{no sum over } k),$<br>$\text{shape}(T_{\text{full}}) = (n, n, n), \quad \text{shape}(T_{i < j < k}) = \left( \binom{n}{3} \right).$                                                                                                                                                                              | i,j,k are local retained indices. Repeated k is not summed. Saved, unused by Run 1.   |
| 18 P/F features                 | Selected P_L,F_L; canonical AO order                | D133-136; R218-241                                         | Saved diagonal and pair features use the producer ordering. The loader validates them and rebuilds canonical density and Fock matrices. FE and FE-prime use ascending eigenvalues of the selected blocks; density is dimensionless and Fock quantities are in Hartree.    | Diagonals, pairs, full blocks, spectra     | Active Lowdin AO / features | $P_i = (P_L)_{v_i v_i}, \quad F_i = (F_L)_{v_i v_i}, \quad P_{ij} = (P_L)_{v_i v_j}, \quad i < j,$<br>$x_{\text{FE}} = \text{eigvalsh}(P_L^{\text{sel}}), \quad x_{\text{FE\_prime}} = \text{eigvalsh}(F_L^{\text{sel}}).$                                                                                                                                             | FE: density spectrum. FE_prime: Fock spectrum. Ascending eigenvalues; n entries each. |

| Layer                   | Input                                                                         | Exact method / function                                              | What is done                                                                                                                                                                                                                                                                                         | Output                                | Basis                      | Formula                                                                                                                                                                                                                                                                                                                                                                                                  | Important element definition in formula                                                   |
|-------------------------|-------------------------------------------------------------------------------|----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| 19 Population           | 30 accepted records per molecule                                              | R load_population244-282; retained_positions111-117                  | Exactly 30 records per molecule, with IDs 001 through 030. Incomplete or erroneous records are rejected; there is no cutoff filtering. Traversal positions are zero based.                                                                                                                           | Sorted complete population            | Classical records          | sort key = ( $q_A$ , geometry index).                                                                                                                                                                                                                                                                                                                                                                    | $q\_A$ : stored scan coordinate. Sort tie breaker is geometry index; no cutoff filtering. |
| 20 Split                | Population; split manifest or 15-15 selection                                 | R validate_split_manifest41-73;244-269;331-366                       | The default manifest requires explicit disjoint train, validation and test sets covering every geometry. In the explicit 15-15 protocol, membership follows geometry ID rather than sorted traversal position.                                                                                       | Train / validation / test membership  | Classical records          | $\mathcal{I}_{\text{train}} = \{1, 3, \dots, 29\}, \quad \mathcal{I}_{\text{test}} = \{2, 4, \dots, 30\} \quad (15-15).$                                                                                                                                                                                                                                                                                 | Geometry IDs are one based. Odd/even membership does not refer to traversal positions.    |
| 21 Diagonal scaling     | Training diagonal entries for each descriptor type                            | C compute_encoding_constants345-392; standardize_diagonal394-399     | MB scaling pools every training geometry and selected AO, separately for density and Fock diagonals. Population standard deviation is used. Positive finite standard deviations are required; there is no epsilon floor or clipping.                                                                 | Scalar mean/std; standardized entries | Classical features         | $K = N_{\text{train}}n, \quad \mu_x = \frac{1}{K} \sum_{a=1}^K x_a, \quad \sigma_x = \sqrt{\frac{1}{K} \sum_{a=1}^K (x_a - \mu_x)^2},$<br>$x_a^{\text{std}} = \frac{x_a - \mu_x}{\sigma_x} \quad (\text{ddof} = 0).$                                                                                                                                                                                     | K counts all training geometry-AO entries. Separate statistics for P and F; ddof=0.       |
| 22 Pair/triple indexing | Selected local AO count n                                                     | C pair_addresses241-244; triple_addresses245-248;249-262             | Local pairs follow row-major strict-upper ordering; triples follow nested lexicographic ordering. Pair values feed scale fitting. The triple helper is unused by Run 1.                                                                                                                              | Ordered pair/triple addresses         | Local AO indices           | $0 \leq i < j < n, \quad N_{\text{pair}} = \binom{n}{2}; \quad 0 \leq i < j < k < n, \quad N_{\text{triple}} = \binom{n}{3}.$                                                                                                                                                                                                                                                                            | Pairs i<j; triples i<j<k. No global AO numbering in these enumerations.                   |
| 23 Pair/triple scales   | Pooled training absolute pair/triple values                                   | C _p95_abs279-287;381-392                                            | The percentile uses linear interpolation of pooled absolute entries, including zeros. Empty or nonpositive scales reject. MB-1 fits the pair scale even though it has no pair gates. The triple scale belongs only to the MB-3 helper.                                                               | alpha; optional helper beta           | Classical scale parameters | $q_{95} = \text{percentile}_{95}( x ), \quad \alpha = \frac{\text{atanh}(0.9)}{q_{95}}.$<br>$\beta$ is fitted analogously for triple inputs.                                                                                                                                                                                                                                                             | Percentile includes zeros and uses linear interpolation. Positive scale is mandatory.     |
| 24 Local encoding       | MB standardized diagonals and a,b,c; or raw fingerprint and fitted statistics | C CachedCircuitBank461-466; encode755-763; FingerprintCircuit800-820 | MB shares three trainable encoding scalars across wires. Fingerprints instead use training-only per-column statistics. For fingerprint columns, replace near-zero standard deviations by one, explicitly zero those columns after scaling, and then clip. RY uses the library half-angle convention. | Dimensionless rotation angles         | Features to circuit wires  | $\theta_i^{\text{MB}} = \frac{\pi}{2} \tanh(aP_i^{\text{std}} + bF_i^{\text{std}} + c),$<br>$m_j = [\sigma_j < 10^{-12}], \quad \sigma_j^{\text{safc}} = \begin{cases} 1 & m_j, \\ \sigma_j & \text{otherwise,} \end{cases}$<br>$u_j = \begin{cases} 0 & m_j, \\ \frac{\pi}{2} \frac{x_j - \mu_j}{\sigma_j^{\text{safc}}} & \text{otherwise,} \end{cases} \quad \theta_j = \text{clip}(u_j, -\pi, \pi).$ | a,b,c shared across MB wires. Fingerprint mask is per column; clipping is last.           |
| 25 Pair encoding        | Pair entries and fitted alpha (helper)                                        | C467-473,610-640                                                     | Pair encoding exists for helper models. MB-1 supplies no pair inputs and queues no pair gates; this is not an active Run 1 stage.                                                                                                                                                                    | Pair gates; none for MB-1             | Circuit / wire             | $\text{IsingZZ}\left(\frac{\pi}{2} \tanh(\alpha P_{ij})\right), \quad i < j.$                                                                                                                                                                                                                                                                                                                            | IsingZZ library gate; local i<j. Not executed by the launched MB-1 branch.                |
| 26 Triple encoding      | Triple entries and fitted beta (helper)                                       | C474-481                                                             | Triple encoding exists as a helper. No current Run 1 method supplies triple inputs, and triple-model runtime results were not numerically verified.                                                                                                                                                  | Triple gates; none in Run 1           | Circuit / wire             | $\text{MultiRZ}\left(\frac{\pi}{2} \tanh(\beta T_{ijk})\right), \quad i < j < k.$                                                                                                                                                                                                                                                                                                                        | MultiRZ library gate; local i<j<k. Runtime not verified for this helper.                  |
| 27 HEA                  | Encoded state; shared layer parameters phi                                    | C shared_ansatz416-424                                               | There is a single encoding followed by two shared-parameter ansatz layers. Each layer applies the same RY angle to all wires, then sequential ring CNOTs in ascending wire order, with the wrap edge last. For two wires, both directed edges execute.                                               | Two-layer ansatz state                | Circuit / parameter        | $L = 2: \quad RY_q(\phi_\ell) \text{ for every } q; \quad \text{CNOT}_{q \rightarrow (q+1) \bmod n}, \quad q = 0, \dots, n-1.$                                                                                                                                                                                                                                                                           | One phi per layer, shared across wires. CNOT order is sequential, wrap edge last.         |

| Layer               | Input                                                                  | Exact method / function                                      | What is done                                                                                                                                                                                                                                                                                                                                                                  | Output                              | Basis                     | Formula                                                                                                                                                                                                                                                                     | Important element definition in formula                                                        |
|---------------------|------------------------------------------------------------------------|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| 28 Measurement      | Final circuit state                                                    | C482-483,818-820                                             | Measure Pauli-Z on every physical wire in wire order. The output is dimensionless. Simulation and autodifferentiation are delegated; independent statevector predictions were checked, but no separate direct-runtime Z-vector residual was recorded.                                                                                                                         | z: n entries                        | Wire expectations         | $z_q = \langle \psi   Z_q   \psi \rangle, \quad z \in \mathbb{R}^n.$                                                                                                                                                                                                        | q=0,...,n-1, physical wire order. Pauli-Z is dimensionless.                                    |
| 29 Readout features | Unpadded z; selected wire count n                                      | C pad_z_to_length568-574; invariant_moments576-583;641-660   | Zero padding adds classical features, not wires. Moment aggregation uses the unpadded measurement vector.                                                                                                                                                                                                                                                                     | f: m+3 entries                      | Classical features        | $m = \max(8, n), \quad z_{\text{pad}} = (z_0, \dots, z_{n-1}, 0, \dots, 0) \in \mathbb{R}^m,$<br>$M_k = \frac{1}{n} \sum_{q=0}^{n-1} z_q^k, \quad k = 1, 2, 3, \quad f = (z_{\text{pad}}, M_1, M_2, M_3) \in \mathbb{R}^{m+3}.$                                             | m=max(8,n). Moments use n measured entries, not the padded vector.                             |
| 30 Prediction       | f; trainable readout weights and bias                                  | C CorrelationEnergyModel655-681; FingerprintModel880-910     | The scalar prediction is in Hartree; targets are not standardized. For fingerprint models, the number of wires is the fingerprint width rather than necessarily the AO count.                                                                                                                                                                                                 | Scalar predicted correlation energy | Parameter / scalar energy | $\hat{y} = b + \sum_{j=1}^{m+3} w_j f_j.$                                                                                                                                                                                                                                   | w has m+3 entries; b is readout bias, distinct from MB encoding coefficient b. Ha.             |
| 31 Initialization   | Seed; model layout; training targets                                   | T initialize_parameters18-32; C843-848                       | Initialization uses a local seeded CPU Torch generator. MB draws one complete vector; fingerprints draw separate ansatz, bias and weight blocks. The bias is then overwritten by the mean training target.                                                                                                                                                                    | Initial model parameters            | Parameter space           | $\theta^{(0)} \sim 0.05 \mathcal{N}(0, I), \quad b^{(0)} = \frac{1}{N_{\text{train}}} \sum_{a \in \text{train}} y_a,$<br>$D_{\text{MB}} = L + m + 7, \quad D_{\text{fingerprint}} = L + m + 4.$                                                                             | D is total parameter count. Gaussian draws occur before target-mean bias overwrite.            |
| 32 Batches/loss     | Training set; persistent shuffle generator; batch size                 | T SampleOrder34-43; iter_batches45-52; mse_loss70-73;183-233 | An independently maintained, same-seed shuffle generator produces a fresh permutation each epoch. Keep the final partial batch. Loss is in squared Hartree and divides by the actual batch size.                                                                                                                                                                              | Batches; scalar MSE                 | Classical / parameter     | $\mathcal{L}_B = \frac{1}{ B } \sum_{a \in B} (\hat{y}_a - y_a)^2.$                                                                                                                                                                                                         | B is actual batch, including final partial batch. Loss unit Ha squared.                        |
| 33 Adam             | Loss gradients; bound parameter blocks; optimizer state                | T54-91; C850-873                                             | Each update clears gradients to None, differentiates, binds gradients according to the model parameter layout, and steps Torch Adam. MB uses one vector group; fingerprints use shared-storage blocks. Installed defaults and two diagnostic steps were checked; the full training trajectory was not replayed.                                                               | Updated parameters                  | Parameter space           | No additional repository formula; see operation and boundary.                                                                                                                                                                                                               | Torch Adam boundary. Gradient clearing and model-specific parameter storage inspected.         |
| 34 Checkpoint       | Epoch metrics; current theta; selection protocol                       | T BestCheckpoint132-173;211-233; R402-453                    | Checkpoint on strict improvement, retaining the earliest tie. Manifest training selects by validation MAE; 15-15 selects by training MAE. Restore parameters only, not optimizer state, before final test evaluation.                                                                                                                                                         | Best theta; final evaluation        | Parameter space           | $e_* = \min_e \left( \arg \min_e \text{MAE}_{\text{selection}}(e) \right).$                                                                                                                                                                                                 | Strict improvement retains first tie. Optimizer state is not restored.                         |
| 35 Metrics          | Predictions and targets in one split                                   | T error_metrics244-254; evaluate256-266                      | Compute each split separately. Convert scalar metrics from Hartree to milli-Hartree only after reduction.                                                                                                                                                                                                                                                                     | MAE, RMSE, MaxAE, NPE               | Scalar statistics         | $e_a = \hat{y}_a - y_a, \quad \text{MAE} = \frac{1}{N} \sum_a  e_a , \quad \text{RMSE} = \sqrt{\frac{1}{N} \sum_a e_a^2},$<br>$\text{MaxAE} = \max_a  e_a , \quad \text{NPE} = \max_a e_a - \min_a e_a, \quad \text{metric}_{\text{mHa}} = 1000 \text{metric}_{\text{Ha}}.$ | e is signed prediction minus target. Convert after reduction: Ha to mHa.                       |
| 36 Seed summaries   | Per-seed, per-molecule metrics                                         | T compute_statistics268-272; R summarize455-480; P131-149    | The full launcher requires 32 seeds. For overall results, first average molecules with equal weight within each seed, then summarize seeds. Also report the median and count. Core singleton SD/SE are NaN; the plot helper uses SD zero and SE NaN.                                                                                                                          | Mean, SD, median, count, SE         | Classical aggregation     | $\bar{x} = \frac{1}{S} \sum_{s=1}^S x_s, \quad s_x = \sqrt{\frac{1}{S-1} \sum_{s=1}^S (x_s - \bar{x})^2}, \quad \text{SE} = s_x / \sqrt{S}.$<br>$x_s^{\text{overall}} = \frac{1}{N_{\text{mol}}} \sum_m x_{m,s}.$                                                           | S is seed count here, not overlap. Sample SD uses ddof=1. Molecules equally weighted.          |
| 37 Plot assembly    | Configured seeds; predictions; initial-data arrays for separate branch | P validate_seed_runs152-168; load_prediction346-375;377-397  | Plotting checks exact configured seed coverage and equality of sorted geometry ID, split, bond length and target. Bars show test mean plus/minus SE; curves show per-geometry signed-error mean plus/minus sample SD. FE-prime is absent from the plotting CLI. Initial-data plots are a separate branch, with 31 curve points including equilibrium and 30 scan rows in CSV. | Error plots; energy curves          | Classical plotting        | $E_{\text{relative}} = 1000 [E - E_{\text{RHF}}(r_e)] \quad \text{mHa}; \quad E_{\text{absolute}} = E \quad \text{Ha}.$                                                                                                                                                     | Seed identity includes geometry ID, split, bond length and target. Curve and CSV sizes differ. |